MILITARY SPECIFICATION

IGNITION MATERIAL, PELLETED, PYROTECHNIC FOR IGNITER MARK 265

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification defines the minimum requirements for one type of pelleted pyrotechnic ignition material.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129 Marking for Shipment and Storage

DRAWINGS

Bureau of Naval Weapons, Department of the Navy

2516382 Charge, Igniter Assembly

(When requesting any of the above documents, give the title and complete designation of the item as shown above. Copies of this specification and other unclassified specifications, standards, and publications required by contractor in connection with specific procurement functions may be obtained from the Commanding Officer, Naval Supply Depot (Code 105), 5801 Tabor Avenue, Philadelphia 20, Pennsylvania. Drawings may be obtained from the Commanding Officer, Naval Ordnance Plant (CTDO), Iroquois Station, Louisville, Kentucky. Contractors should requisition forms from the cognizant field contract administrator. Field contract administrators should obtain BuWeps Instructions from the Forms and Publications Segment of the Navy Supply System in accordance with NAVSANDA Publication 2002. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

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PUBLICATION

Interstate Commerce Commission

49 CFR 71-90 Code of Federal Regulations

(Request copies from the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Preproduction. -The preproduction test sample shall be manufactured using the process methods and equipment proposed for production. Any significant changes after preproduction approval must be approved by the procuring activity prior to incorporation into production. It shall be the responsibility of the contractor to inform the procuring activity of proposed changes after approval of the preproduction sample. In the event of significant changes in the process methods and equipment which in the opinion of the procuring activity may adversely affect the characteristics of the material, additional preproduction samples may be required.

3.2 Composition. -The chemical composition of the pellets shall be in accordance with BuWeps Drawing No. 2516382 and shall be fabricated utilizing only the ingredients and weight percentages stated on the drawing.

3.3 Physical Properties. -The physical properties of the pellets shall be in accordance with Table I.

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TABLE I

PHYSICAL PROPERTIES

Characteristics	Limits Minimum	Maximum
Calorimetric Value (Average)	1500 Calories per gram	
Crush Strength (Average)	6500 grams	
Dimensions: Length	0.177 inch	0.197 inch
Diameter	0.121 inch	0.129 inch
Pellet Density (Average)	1.63 grams per cubic cm.	1.77 grams per cubic cm.
Pellet Weight (Average)	0.060 grams	

3.4 Workmanship. -The material furnished under this specification shall be a uniform product free from foreign materials. It shall be uniform in quality and manufactured in accordance with standard manufacturing procedures of the industry.

3.5 Data Requirements. -No data is required by this specification or by applicable documents referenced in section 2 unless specified in the contract or order (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Tests.

(a) Preproduction Tests (4.4)

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(b) Quality conformance inspection (4.5)

4.3 Lots.

4.3.1 Sublot Size. -A sublot of ignition pellets shall consist of a maximum weight of 10 pounds and shall be made with one pelleting press during an 8-hour shift. If more than 10 pounds of pellets are produced in 8 hours, then two or more sublots shall be made up. Ignition pellet production from each press shall be kept separate.

4.3.2 Lot Designation. -Beginning with the first sublot of ignition pellets produced under a specific contract or order, the lots shall be numerically identified by master and sublot designations. A change in the master lot number shall occur any time that a new batch of any chemical component is employed and the sublot designation numbering shall be started all over.

4.4 Preproduction Tests. -Preproduction tests shall consist of all of the Quality Conformance Inspections and requirements of this specification.

4.5 Quality Conformance Inspections. -The following procedures shall be used for determining that the requirements of this specification have been met.

4.5.1 Sampling. -A minimum of three grams of the ignition pellets shall be drawn from each of five widely-separated places from each sublot produced. The entire sample except 35 ignition pellets shall then be cut, shaved, or otherwise reduced in size and thoroughly mixed such that any sample withdrawn for acceptance tests will be representative of the entire sample. The 35 pellets shall be used for the determination of density.

4.5.2 Chemical Composition. -The vendor is required to certify that the chemical composition of the pellets furnished are in accordance with BuWeps Drawing No. 2516382 (See 6.2).

4.5.3 Average Calorimetric Value. -The calorimetric value for the heat of reaction of the ignition pellet shall be determined in duplicate on each sublot using a Parr bomb calorimeter No. 1200 or equal. A charge of 12 to 18 pellets shall be placed in the bomb and the system purged of any residual air using nitrogen, argon, or helium gas. The bomb shall be closed after purging so that an atmosphere

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of inert gas is maintained in the bomb at atmospheric pressure. The pellets shall then be burned and the heat of reaction determined. Failure to comply with requirements of Table I shall be cause for rejection of the sublot.

4.5.4 Average Crush Strength. -The load required to crush the ignition pellet longitudinally shall be determined in duplicate on 10 to 15 pellets from each sublot. This test shall be made using F. J. Stokes Machine Company pellet hardness tester or equal. The load shall be applied in 1000 gram increments along the longitudinal axis and shall be maintained for a minimum of five seconds before increasing the load. Failure to comply with requirements of Table I shall be cause for rejection of the sublot.

4.5.4.1 Alternate Procedure for Determining Crush Strength. -Test pellets on longitudinal axis in a compression tester using a maximum loading rate of 0.4 inch per minute.

4.5.5 Dimensions. -The length and diameter of ten randomly selected pellets shall be checked. Failure of the pellets to meet the requirements of Table I shall be cause for rejection of the sublot.

4.5.6 Average Ignition Pellet Weight and Density. -The average weight test shall be run in duplicate for each sublot. Two groups of 15 to 20 pellets each shall be weighed and then checked for density by such approved means as (a) determination of loss of weight in a liquid medium or (b) determination of weight and volume of liquid displacement. Castor oil is required as the liquid medium.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. -Preservation and packaging shall be level C or as specified (see 6.2).

5.1.1 Level C. -Preservation and packaging shall be in accordance with standard commercial practice to afford protection against damage, contamination of the product by moisture, corrosion, or other foreign matter.

5.2 Packing. -Packing shall be level C or as specified (see 6.2).

5.2.1 Level C. -The product shall be shipped in commercial type containers complying with ICC Code of Federal Regulations 49 CFR 71-90, and so constructed to withstand normal transportation and handling abuse and must be suitably sealed to prevent contamination of the contents by moisture or foreign matter.

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5.3 Marking. -Unless otherwise specified by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 and shall include but not be limited to the following (see 6.2):

(a) Manufacturer's name and address

(b) Gross and net weights

(c) Lot number and date of manufacture

(d) The number of this specification

(e) Safety precautions or unusual storage requirements

(f) Purchase order or contract number

6. NOTES

6.1 Intended Use. -The material covered by this specification is intended for use in the igniter Mk 265.

6.2 Ordering Data. -Procurement documents shall specify but not be limited to the following information:

(a) Title, number, and date of this specification

(b) Type and size of container desired

(c) Place of inspection

(d) Minimum lot size, if applicable

(e) Place of delivery

(f) Request for copies of inspection data, if source inspected

(g) Selection of applicable levels of packing and packaging (see 5.1 and 5.2)

(h) Marking requirements (see 5.3)

(i) Data required (see 4.5.2)

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6.3 Conflicting Requirements. -Conflicting requirements arising between this specification or any specifications, publications or drawings listed herein, shall be referred in writing to the procuring activity or appointed agent for interpretation and clarification.

6.4 Request for Deviation. -Request for deviation from this or referenced specifications, materials, applicable drawings, or processes shall be forwarded to the procuring activity prior to manufacture of the affected item or items. All deviations shall be limited to the contract under which they were granted.

6.5 Request for Waiver. -Request for waiver from this specification, applicable drawings, specifications, materials, or processes shall be forwarded to the procuring activity. Waivers shall be limited to the specific units for which they are granted. A waiver request shall include the following:

(a) Exact nature of defect

(b) CD number and defect classification

(c) Quantity of items involved

6.6 Value Engineering. -All manufacturers of units covered by this specification are encouraged to submit value engineering proposals to the procuring activity in an effort to reduce cost.

6.7 Safety Precautions. -This is a potentially dangerous product. All detailed safety precautions shall be strictly observed as set forth by the manufacturer and user.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-I-23951(Wep) IGNITION MATERIAL, PELLETED, PYROTECHNIC FOR IGNITER MK 265

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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